

Section C - Descriptions and Specifications

PERFORMANCE WORK STATEMENT**Performance Work Statement (PWS)****WS-26-0001 – Initial Task Order****Three-Dimensional High Speed-High Altitude Mapping Operations
& Technology Integration (3DH2-M O&TI)****TBD****Part 1. General Information**

1.1 Introduction and Background: The US Army Geospatial Center's (AGC) Three-Dimensional High Speed-High Altitude Mapping Operations & Technology Integration (3DH2-M O&TI) program provides timely, accurate and relevant High Resolution Three-Dimensional (HR3D) Data to enable geospatial information, capabilities, and domain expertise for the Army Geospatial Enterprise implementation and Geospatial Data Fabric. The 3DH2-M mission provides critical information to the Warfighter in all Domains and in all Combatant Commands (CCMDs). The purpose of the 3DH2-M mission, is to collect, process, disseminate, store, and maintain data collected by conducting wide area high altitude flight operations. Processed 3DH2-M data serves as foundational geospatial data products for the Department of Defense (DoD), Coalition Partners, and Host Nations. Data collection is primarily Outside of the Continental U.S. (OCONUS) with some in the Continental U.S. (CONUS). The data provided from 3DH2-M supports Government efforts for terrain analysis, change detection, threat analysis, intelligence preparation of the battlefield (IPB), operational and tactical mission planning, intelligence, surveillance, and reconnaissance (ISR) support, engineer planning, Aerial Port of Debarkation (APOD), Seaport of Debarkation (SPOD), Large Scale Combat Operations (LSCO), Counter Insurgency (COIN), Humanitarian and Disaster Relief (HADR) and many other uses. The Warfighter Support Directorate (WSD), 3DH2-M Program is a part of the US Army Corps of Engineers (USACE) at the US Army Geospatial Center (AGC). The 3DH2-M Program focuses on providing high fidelity geospatial foundational data/terrain mapping, and provides support to all authorized Army, DoD, and other government organizations.

This Task Order performance work statement (PWS) describes the services and tasks to be undertaken. The purpose is to collect, process, disseminate, apply, and maintain HR3D geospatial data by conducting flight and collection operations and by processing and producing the required 3DH2-M data products (comprised of but not limited to terrestrial and bathymetric LiDAR, EO, Synthetic Aperture Radar (SAR), Hyperspectral Imaging (HSI), Full Motion Video (FMV), Special Intelligence (SI), and unclassified commercial satellite imagery data). In addition, the Department of Defense (DoD) is pursuing advancements in terrestrial and bathymetric LiDAR, SAR, HSI, and FMV sensor technology. This is a result of the operational requirement to provide high fidelity terrain mapping from a higher altitude, higher velocity aircraft platform (to reduce risk from adversary fire). DoD requires advancement in supporting terrestrial and bathymetric LiDAR, SAR, HSI, and FMV data processing, storage, and dissemination technology. Current terrestrial and bathymetric LiDAR collection systems are rapidly improving/increasing area collection rates as well as increasing the geospatial fidelity of the data. In order to keep pace with the improving collection systems, improved, more rapid data processing needs to be integrated into the 3DH2-M program. The 3DH2-M program is also looking for emerging technology integration for LiDAR (terrestrial and bathymetric), EO, SAR, HSI, FMV, SI, and Measurement and Signals Intelligence (MASINT) sensors.

1.2 Scope: This PWS defines the requirements for Contractor-provided services necessary to

support the 3DH2-M program. These services cover all requirements, including the following:

- ▶ Manned fixed wing platform collection of wide area airborne and bathymetric collection that capture 5-, 10- and 50-centimeter spatial resolution EO imagery, 50cm LiDAR, Foliage Penetration (FOPEN) and 20m depth bathymetry data and multi-Intelligence (INT).
- ▶ Provide large scale, worldwide on site, and advanced cloud based geospatial data processing, exploitation, analysis, meshing of data to create and deliver 2, 2.5 and 3D content, and rapid dissemination.
- ▶ Provide a centralized CONUS based **GEOINT** content management, exploitation, analysis and reach back architecture to support end user requests.
- ▶ Provide personnel and equipment to enable **SI data pathways** for exploitation, analysis and reach back architecture to support end user requests.
- ▶ Provide a Contractor Owned, Contractor Operated (COCO) fleet integrated with Government furnished sensors with responsibility for aircraft and sensor maintenance, materials, aircraft and personnel mobilization and demobilization.
- ▶ Provide technology integration and value engineering with an emphasis toward next generation unmanned and terrestrial capabilities with services on demand, as well as rapid data processing workflows with advanced software capabilities and/or Artificial Intelligence/Machine Learning (AI/ML).
- ▶ Provide meetings, deliverables, schedules, configuration management, personnel management, risk management, quality control, security, and overall program compliance.

1.3 Description of Services: The Contractor shall provide a fully integrated flight operation, collection, and processing service to support worldwide 3DH2-M collections requirements. The Contractor shall provide all personnel, equipment, supplies, facilities, transportation, tools, materials, supervision, and other items necessary to perform the 3DH2-M support tasks as defined in this PWS, except for those items specified as Government-furnished property (GFP), services and information.

1.4 Objectives: The objective of this effort is to collect, process, disseminate, exploit, store, and manage airborne terrestrial and bathymetric LiDAR, EO, SAR, HSI, Special Intelligence (SI), multi- Intelligence (INT) to support the end user 3DH2-M requirements and operations.

Period and Place of Performance: This task order will have one (1) base year and one (1) option year. Place of performance is at the OCONUS base camp locations identified in PWS paragraph 5.2.1 and within the designated AOR's listed for each base camp, and the CONUS locations identified in paragraphs 5.2.1.2 and 5.2.1.3. Some locations will be utilized as directed by the Government to support operations within an AOR.

Access to and Disclosure of Information: The Contractor may require access to data and information proprietary to a government agency, another Government Contractor, or of such nature that its dissemination or use, other than as specified in this contract, would be adverse to the interests of the Government or others. Except to authorized Government personnel, or upon written approval of the Contracting Officer or designated representative, neither the Contractor, nor Contractor employees, shall divulge nor release data or information developed or obtained in the performance of this contract.

The Contractor shall direct to the Program Manager (PM) and Contracting Officer's Representative (COR) all inquiries, comments, or complaints arising from matters observed,

experienced, or learned as a result of, or in connection with the performance of this contract, the resolution of which may require the dissemination of official information.

Inquiries received by the Contractor regarding work performed under this contract shall be referred to the Contracting Officer for evaluation under the Freedom of Information Act (FOIA) of 1975, Public Law 93-502, 5 U.S.C., Section 552. The Contractor shall not release any information concerning this project to the public without written authorization from the Contracting Officer.

1.5 Quality Control: The Contractor shall implement, maintain, and use Quality Control (QC) and quality assurance (QA) procedures to ensure proper oversight for data collected, personnel, training, facilities, equipment, supplies, services, and subcontractors. The Contractor's Quality Control Plan (QCP) is the means by which the Contractor assures its work complies with the requirements of the contract. The Contractor shall develop and maintain an effective QCP to ensure services are satisfactorily performed under this contract. The Contractor shall implement procedures to identify, prevent, and ensure non-recurrence of defective services and data products. The Contractor's QCP is the means by which the Contractor shall develop QCP procedures that address the areas identified in Section 5, Specific Tasks in this PWS.

The QCP shall address the internal quality control processes with clear and direct lines of communication to ensure accountability to the Contractor's senior management and leadership; address overall project management and administration; cover all services described in this PWS to include essential services as identified in the Performance Requirements Summary (PRS); describe a method, acceptable to the Government, of identifying deficiencies in the quality of service performed under this contract before the level of performance becomes unacceptable, and address processes for corrective actions without dependence upon Government direction; require documentation of all Contractor QC inspections and corrective actions be maintained by the Contractor throughout the term of this contract. Copies of inspection documents will be provided to the COR upon request and they will describe the methods of communication with the Government regarding performance of the contract. The QCP shall include regular and formal meetings with the Government and proposed formats and/or schedules for informal communications (e.g., periodic emails to the PM and COR).

1.6 Quality Assurance: The Government will evaluate the Contractor's performance under this contract in accordance with the QCP and the Government's Quality Assurance Surveillance Plan (QASP). The QASP will be used to evaluate contractor performance in accordance with the performance standards. It defines how the performance standards will be applied, the frequency of surveillance, and the minimum acceptable defect rates.

1.7 Government Remedies: The Contracting Officer will follow Federal Acquisition Regulation (FAR) 52.246-5, "Inspection of Services - Cost Reimbursement" for the Contractor's failure to perform satisfactory services or failure to correct nonconforming services.

1.8 Recognized Holidays: All federal holidays are to be observed by the Contractor personnel performing within CONUS.

Hours of Operation: For OCONUS operations and support under this effort, normal operating hours will be 24 hours a day, 7 days a week (24/7); and individual employees shall be expected to work 12 hours a day, 7 days a week to support the 24/7 operations. For the portions of this effort that are conducted CONUS, the Contractor shall be responsible for conducting business between the hours of 9:00am and 5:00pm, Monday through Friday, except for supporting emergency management operations during which time business may need to be conducted 24 hours a day for the duration of the emergency management operations.

1.9 Security Requirements

All security requirements are addressed in the DD Form 254. Key Contractor Personnel (see Section 5.6.4 of this PWS), with the exception of the Chief Mechanic, will possess Top Secret (TS) Sensitive Compartmented Information (SCI) security clearances to work on the contract. The Chief Mechanic and non-Key Contractor personnel that are not deployers, performing work onsite at any of the various processing locations will be required to have a security clearance of no lower than Top Secret. All deployed personnel shall have a security clearance of no lower than Secret to work on the contract.

The Contractor shall maintain at least a TS Facility Clearance and make available approved Secret storage sufficient to accommodate the needs of this PWS.

Physical Security: The Contractor shall be responsible for safeguarding all Government equipment, information, and property provided to the Contractor.

1.10 Antiterrorism/Operational Security (AT/OPSEC)

1.10.1 The security requirements described below apply to all contract personnel (including employees of the prime Contractor ("Contractor") and all subcontractor employees) supporting the performance requirements of this contract. The Contractor is responsible for compliance with these security requirements. Questions regarding security matters shall be addressed to the designated Government representative (e.g., Contracting Officer Representative (COR), Requiring Activity (RA) representative, or Contracting Officer (if a COR or other RA representative is not appointed). Contract personnel are critical to the overall security and safety of US Army Corps of Engineers (USACE) installations, facilities and activities, and security awareness training. The Department of Defense (DoD) and Army security training requirements specified below, if applicable, are performance requirements; all applicable contract personnel shall complete initial training within 30 days of contract award or the date new contract personnel begin performance on the contract. Within five business days from the completion of training, the Contractor shall provide written documentation (e.g., email or memorandum) to the Government representative. The documentation shall include the names of contract personnel trained and which training they completed; the Contractor shall maintain training records as part of their contract files and be prepared to provide copies of training certificates to the Government representative. Contractor personnel and vehicles are subject to search when entering federal installations. Additionally, all contract personnel shall comply with Force Protection Condition (FPCON) measures, Random Antiterrorism Measures (commonly referred to as "RAMs"), and Health Protection Condition (HPCON) measures. The Contractor is responsible for meeting performance requirements during elevated FPCON and/or HPCON levels in accordance with applicable RA plans and procedures --this includes identifying mission essential and non-mission essential personnel. In addition to the changes otherwise authorized by the changes clause of this contract, should the FPCON or HPCON levels at any individual facility or installation change, the Government may implement security changes that affect contract personnel. The Contractor shall ensure all contract personnel are aware of their security responsibilities, including any site-specific requirements identified in local policies or procedures.

1.10.2 All contract personnel requiring routine access to Army installations, facilities, and controlled access areas, or requiring network access shall complete initial and annual refresher AT Level I awareness training. Online AT Level I awareness training is available at <https://jko.jten.mil/> (website subject to change).

1.10.3 All contract personnel requiring physical access to a federal installation or facility shall comply with the access control procedures of that location. Contract personnel requiring unescorted access to meet contract performance requirements on a DoD installation in the US shall be vetted by the installation/facility Provost Marshal/Directorate of Emergency Services/Security Office using the National Crime Information Center-Interstate Identification Index (commonly referred to as "NCIC-111") and Terrorist Screening Database (commonly referred to as "TSDB"). Contract personnel shall comply with all personal identity verification requirements specified in installation/facility policies and procedures. Contract personnel who do not meet requirements for unescorted access to USACE facilities shall coordinate escorted access with the Government representative, as needed. Contract personnel who receive keys, access cards, or lock combinations that provide access to government-owned property shall comply with key and lock control procedures of the RA.

1.10.4 Contract personnel will be issued a common access card (CAC) only if duties involve one of the following: (1) both physical access to a DoD facility and access to DoD information systems or networks; (2) remote access to a DoD information system or network using DoD-approved remote access procedures; or (3) physical access to multiple DoD facilities or multiple non-DoD federally controlled facilities on behalf of the DoD on a recurring basis for a period of 6 months or more. Before CAC issuance, contract personnel must have, at a minimum, a favorably adjudicated Tier 1 investigation or an equivalent or higher investigation in accordance with applicable Army regulations and Homeland Security Presidential Directive-12 (HSPD-12). At the discretion of the RA, an initial CAC may be issued based on a favorable review of a fingerprint check and a successfully scheduled Tier 1 investigation with the National Background Investigations Bureau. The RA provides contract personnel with additional information and forms to initiate the CAC issuance process, and/or to initiate background investigations, when required. Contract personnel shall complete these processes within established timelines to avoid delays.

1.10.5 For contract performance requirements that involve services or delivery in a foreign country, the Contractor shall comply with the requirements of DFARS clause 252.225-7043. For performance requirements that involve contract personnel accompanying or supporting US Armed Forces deployed outside the US, the Contractor shall comply with the requirements of DFARS clause 252.225-7040. Contract personnel accessing DoD or other federal facilities outside the US shall comply with applicable Status of Forces Agreements and Geographic Combatant Command requirements. Prior to contract personnel traveling outside the US, the Contractor shall provide documentation of AT, OPSEC, and other required training to the Government representative. Additionally, contract personnel shall comply with requirements specified in the DoD Foreign Clearance Guide, including country and theater clearance processes.

1.10.6 All contract personnel shall receive initial and annual refresher training from the RA representative on the local suspicious activity reporting program. This locally developed training provides contract personnel with general information on suspicious behavior, and guidance on reporting suspicious activity to the project manager, security representative or law enforcement entity.

1.10.7 All contract personnel with access to a government information system (including USACE business systems and CAC-enabled websites) shall comply with applicable DoD and Army regulations, and shall use the organization's UserID-Password Administration and Security System (U-PASS) at commencement of services to request network user accounts. Contract personnel shall complete DoD Information Assurance Awareness training prior to accessing information systems, and annually thereafter.

1.10.8 The Contractor shall develop an OPSEC SOP/Plan within 90 days of contract award. The OPSEC SOP/Plan shall be reviewed and accepted by the RA OPSEC Officer. This plan will include a process to identify critical information, where it is located, who is responsible for it, how to protect it, and why it needs to be protected. In accordance with Army and DoD regulations, the contractor shall have a certified Level II OPSEC coordinator, who shall ensure OPSEC compliance during contract performance.

1.10.9 All contract personnel with access to critical information (as identified in the RA's OPSEC Program) shall complete initial and annual refresher OPSEC Level I Awareness training, which is available at the following websites: <https://www.iad.gov/ioss/>, or <http://www.cdse.edu/catalog/operations-security.html> (websites subject to change). All contract personnel with access to Controlled Unclassified Information (CUI) shall complete initial and annual refresher CUI training in accordance with applicable Army policy.

1.10.10 The prime Contractor shall have a Facility Clearance (FCL) at the appropriate level prior to performance on the contract; the RA will sponsor the prime contract company in obtaining the FCL. All cleared contract personnel shall comply with the FCL requirements, as well as applicable laws and regulations regarding contractor access to national security information. For classified contracts, the RA will generate the DD Form 254, which shall be incorporated as an attachment to the contract.

1.10.11 All contract personnel who maintain an active security clearance shall receive initial and annual refresher training on the Threat Awareness and Reporting Program (commonly referred to as "TARP"), provided by a Counterintelligence Agent. As determined by the servicing Counterintelligence Agent for the RA, contract personnel may complete web-based TARP training.

1.10.12 Contractors shall comply with the requirements set forth in FAR clause 52.222-54 Employment Eligibility Verification and FAR Subpart 22.18 in using the E-Verify Program at (<https://www.e-verify.gov/>) (website subject to change) to meet the contract employment eligibility requirements. Contractors are encouraged to cooperate with Federal and State agencies responsible for enforcing labor requirements to include eligibility for employment under United States immigration laws in accordance with FAR 22.102-1(i). An initial list of verified/eligible

candidates shall be provided to the COR no later than three business days after the initial contract award. When contracts are with individuals, the individuals will be required to complete a Form 1-9, Employment Eligibility Verification, and submit it to the Contracting Officer to become part of the official contract file.

1.11 Special Qualifications: Expertise and past performance in wide area high altitude collections and data processing simultaneously in multiple CCMDs.

1.12 Post Award Conference/Periodic Progress Meetings:

The Contractor shall conduct a kickoff meeting two weeks after contract award telephonically, virtually, or in person at the AGC in Alexandria, VA, or at the Contractor's facility.

The Contractor shall conduct program management reviews monthly either by telephone, virtually, in person at the AGC in Alexandria, VA, or in person at the Contractor's facility.

The Contractor shall conduct weekly technical status updates telephonically or virtually with the COR and in-theater Government representatives.

The COR for this contract will be designated upon contract award. A letter of designation shall be issued to the COR, with a copy of provided to the Contractor. The letter of designation, states the responsibilities and limitations of the COR, especially with regard to changes in cost or price, estimates, or changes in delivery dates. The COR is not authorized to change any of the terms and conditions of the resulting contract.

The COR monitors all technical aspects of the contract and assists in contract administration. The COR is authorized to perform the following functions:

- Assure the Contractor performs the technical requirements of the contract;
- Perform inspections necessary in connection with contract performance;
- Maintain written and oral communications with the Contractor concerning technical aspects of the contract;
- Issue written interpretations of technical requirements, including Government drawings, designs, and specifications;
- Monitor Contractor's performance and notify both the Contracting Officer and Contractor of any deficiencies;
- Coordinate availability of GFP; and provide site-entry of Contractor personnel.

Contract Manager: The Contractor shall provide a Contract Manager who, during the post-award phase, will participate in modifications and work to ensure that the services being provided by the Contractor meet the requirements of the PWS. The name of this person and an alternate who shall act for the Contractor when the primary Contract Manager is absent shall be designated in writing to the Contracting Officer. The Contract Manager or alternate shall have full authority to act for the Contractor on all contract matters relating to daily operation of this contract.

Identification of Contractor Employees: All contract personnel attending meetings, answering Government telephones, and working in other situations where their Contractor status is not obvious to third parties are required to identify themselves as such to avoid creating an impression in the minds of members of the public that they are Government officials. They must also ensure all documents or reports produced by Contractors are suitably marked as Contractor

products or that Contractor participation is appropriately disclosed.

1.13 Travel: To the maximum possible extent, the Contractor shall provide electronic media over existing secure high bandwidth or other secure communications links rather than by Contractor travel. The Contractor shall travel between the Contractor's facility and OCONUS locations as required to carry out the tasks specified in this PWS. The Contractor shall comply with the Joint Travel Regulations (JTR). The JTR are applicable to uniformed members of the seven uniformed services (i.e., Uniformed Members of the Army, Navy, Air Force, Marine Corps, Coast Guard, National Oceanic and Atmospheric Administration Commissioned Corps, and Public Health Service Commissioned Corps), DoD civilian employees and others traveling at DoD expense.

All flight crew and aircraft maintenance personnel assigned to perform services under this contract at OCONUS locations shall be subject to a minimum continuous duty period of sixty (60) days at such locations. All other personnel performing services under this contract at OCONUS locations are subject to a minimum ninety (90) day continuous duty period. All personnel performing services under this contract at CONUS locations are subject to a minimum thirty (30) day continuous duty period. This minimum will not apply to CONUS efforts lasting less than 30 days. Exceptions to these minimums must be approved by the Government.

1.14 Documentation: The Contractor shall deliver to the COR all documents and materials as deliverables developed under this effort for inspection and acceptance. The Contractor shall provide all document deliverables in hard copy, softcopy, or secure communications in accordance with the DO1423 CDRLs. The Contractor shall document all work performed under this effort as follows:

1.14.1 The Contractor shall provide monthly reports on the 10th day of each month beginning the month after contract award. The Contractor shall deliver the monthly status reports via secure communications to the COR on the tenth day of each month after contract award. The Contractor shall provide monthly status reports summarizing current program accomplishments measured against the tasks outlined previously, identification and summary of any unresolved problems, incurred costs, meetings attended, planned activities for the next month, issues, and recommendations. The Contractor shall ensure the monthly report includes a spending plan for the period of performance; actual expenditures to date; actual expenditures for the month just ended; and monthly and cumulative variation of actual expenditures from the planned expenditures.

The Contractor shall include in the spending plan all costs to include labor, overhead expenses, description (to include cost and results obtained) of each trip taken in the previous month, and fees. The Contractor shall include detailed copies of all invoices submitted to date for payment from both the prime contractor as well as all subcontractors. The Contractor shall ensure each invoice details the labor categories, labor rates, labor hours, and all ODCs, Materials.

1.14.2 The Contractor shall also answer the following questions within the monthly reports:

- Is the work progressing on schedule?
- What is the percent of completion/against objective?
- What is the dollar value completed/budgeted?
- What is the total number of contract employees currently on contract/objective?
- Was Contractor Manpower Reporting (CMR) performed during the monthly reporting period? If so, what information was entered?
- What are names of the Contractor personnel, who are/were deployed during the period?
- What is the total monthly cost per deployment location/site?
- How many Square Kilometers (SQ/KM) of data was collected, by each aircraft?
- How many SQ/KM of data by each aircraft has been processed and what is the data processing backlog by location?
- How many hours of flight by aircraft were undertaken to collect vs the number of hours the aircraft flew?
- Operational status of each aircraft, next phased maintenance scheduled, and number of hours aircraft is expected to be out of service?

1.14.3 The Contractor shall provide to the COR a weekly status report, to include any incidents of data collection for which data is unusable, corrupted, or indicate error in sensor or sensor operators.

1.14.4 The Contractor shall provide to the COR an accident report and a serious incident and compliance report no later than one (1) day after an accident or serious incident.

1.14.5 The Contractor shall provide to the PM and COR a detailed report of all incidents of collection errors made by sensor operators resulting in unusable data, corrupted data and/or cancelled missions.

1.14.6 Final Report: The Contractor shall submit a draft Final Report one (1) month prior to the end of the contract to the COR for review. The COR will review the draft Final Report within two (2) weeks. The Contractor shall provide a revised Final Report based on COR comments one (1) week prior to the end of the contract.

1.14.7 The Contractor shall submit a Property Control Plan to the COR within one (1) week after contract award.

1.14.8 The Contractor shall submit a Transition Plan with their proposal submission in response to the RFP. The plan must detail obtaining an Air Worthiness Release (AWR) for all aircraft and achieve full operational readiness within 120 days of contract award. The Contractor shall also submit a Contract Close-out Transition Plan no later than 8 months prior to the contract end date, which details how the Contractor can ramp-down and transition to a new contractor, and/or remove and package all GFE, if applicable. The Government will give the Contractor 30 days' notice when the Contract Close-out Transition Plan must be implemented.

1.14.9 The Contractor shall submit an OPSEC plan to the COR within 90 calendar days of award.

1.14.10 The Contractor shall provide daily maintenance reports that adhere to the following:

- The Contractor shall provide daily maintenance reports (by fax or secured email transmission) to the COR and any other TPOC each day not later than 1700 deployment site time.
- The Contractor shall ensure the daily maintenance report contains any adjustments and changes that are made to the system, current airframe and system readiness and logistics issues, upcoming TBO replacements, Programmed Maintenance Schedule, estimated FMC dates for inoperable aircraft or systems, and the status of all deployed personnel and equipment.

Part 2. Definitions and Acronyms

2.1 Definitions

CONTRACTOR

A supplier or vendor awarded a contract to provide specific supplies or service to the Government. The term as used in this contract refers to the prime contractor.

CONTRACTING OFFICER

A person with authority to enter into, administer, and/or terminate contracts and make related determinations and findings on behalf of the Government. Note: The only individual who can legally bind the Government.

CONTRACTING OFFICER'S REPRESENTATIVE (COR)

An individual, including a contracting officer's technical representative (COTR), designated and authorized in writing by the contracting officer to perform specific technical or administrative functions.

DEFECTIVE SERVICE

A service output that does not meet the standard of performance described in the PWS and associated QASP.

DELIVERABLE

Anything that can be physically delivered, including data deliverables in accordance with the CDRLs, and also including non-manufactured things such as meeting minutes or reports.

3DH2-M SYSTEM

An airborne or terrestrial vehicle, equipped with the 3DH2-M sensors, together with the 3DH2-M data processing capability.

KEY PERSONNEL

"Personnel" means employees of the contractor, or any subcontractor(s), affiliates, joint venture partners, or team members, and consultants engaged by any of those entities.

The personnel specified as "Key Personnel" in Section 5.6.4 of this PWS are considered to be essential to the work being performed under this contract. Prior to diverting any of the specified individuals to other projects, the Contractor shall notify the Contracting Officer reasonably in advance and shall submit justification (including proposed substitutions) in

sufficient detail to permit evaluation of the impact on the program. No diversion shall be made by the Contractor without the written consent of the Contracting Officer.

PERFORMANCE WORK STATEMENT

Statement of work (SOW) for performance-based acquisitions that describes the required results in clear, specific, and objective terms with measurable outcomes.

PHYSICAL SECURITY

Actions that prevent the loss or damage of Government property.

PLATFORM

An airborne or terrestrial vehicle.

QUALITY ASSURANCE

Actions taken by the Government to assure contracted services meet PWS requirements.

QUALITY CONTROL PLAN

A written document specifying the surveillance method used for surveillance of Contractor performance.

QUALITY CONTROL

Actions taken by a Contractor to control the performance of contracted services to meet PWS requirements.

SUBCONTRACTOR

One that enters into a contract with a prime Contractor. The Government does not have privity of contract with the subcontractor.

WORKDAY

The number of hours per day the Contractor provides services in accordance with the contract.

WORKWEEK

Monday through Friday, unless specified otherwise.

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2.2 Acronyms

24/7	24 hours a day, 7 days a week
3D	Three Dimensional
AC	Advanced Capability
ACOR	Alternate Contracting Officer's Representative
AFARS	Army Federal Acquisition Regulation Supplement
AFRICOM	U.S. Africa Command
AGC	Army Geospatial Center
AGL	Above Ground Level
ALTM	Airborne Laser Terrain Mapper
AOR	Area of Responsibility
AR	Army Regulation
AT	Anti-Terrorism
AWR	Air Worthiness Release
CAP	Contractor Acquired Property
CCE	Contracting Center of Excellence
CCMD	Combatant Command
CENTCOM	U.S. Central Command
CF	Coalition Forces
CFE	Contractor-Furnished Equipment
CFR	Code of Federal Regulations
CLIN	Contract Line-Item Number
CMR	Contractor Manpower Reporting
CONEX	Container Express
CONUS	Continental United States (excludes Alaska and Hawaii)
COR	Contracting Officer's Representative
COTS	Commercial Off-the-Shelf
CRC	CONUS Replacement Center
CWBS	Contract Work Breakdown Structure
DA	Department of the Army
DCMA	Defense Contract Management Agency
DD	Department of Defense Form
DD250	Department of Defense Form 250 (Receiving Report)
DD254	Department of Defense Contract Security Requirement List
DEM	Digital Elevation Model
DFARS	Defense Federal Acquisition Regulation Supplement
DMDC	Defense Manpower Data Center
DoD	Department of Defense
EC	Enhanced Capability
ELINT	Electronic Intelligence
EO	Electro Optical
EUCOM	U.S. European Command
FAA	Federal Aviation Administration
FAR	Federal Acquisition Regulation
FMC	Fully Mission Capable
FPCON	Force Protection Condition

GCS	Ground Control Station
GeoTIFF	Geographically Tagged Image File Format
GFP	Government-Furnished Property
GFI	Government-Furnished Information
GFR	Government Flight Representative
GOTS	Government Off-the-Shelf
GPS	Global Positioning System
GS	General Services
GSE	Ground Support Equipment
HAZMAT	Hazardous Material
HIPAA	Health Insurance Portability and Accountability Act of 1996
HITL	Hardware in the Loop
HQDA	Headquarters, Department of Army
3DH2-M	Three-Dimensional High Speed-High Altitude Mapping
HR3DGI	High-Resolution Three-Dimensional Geospatial Information
HSI	Hyper-spectral Imaging
IA	Information Assurance
IMS	Integrated Master Schedule
INDO-PACOM	U.S. Indo-Pacific Command
ITAR	International Traffic in Arms Regulation
JHU	John Hopkins University
JP2	JPEG 2000
JPG	Joint Photographic Experts Group
KO	Contracting Officer
LAS	Laser File Format Exchange
LIDAR	Light Detection and Ranging
LOA	Letter of Authorization
MD	Maryland
MFW	Manned Fixed Wing
MOA	Military Operating Area?
MrSID	Multi-resolution Seamless Image Database
N/A	Not Applicable
NGA	National Geospatial-Intelligence Agency
NITF	National Image Transmission Format
NORTHCOM	U.S. Northern Command
OCI	Organizational Conflict of Interest
OCONUS	Outside the Continental United States (includes Alaska and Hawaii)
ODC	Other Direct Costs
OPSEC	Operational Security
PDSS	Pre-Deployment Site Survey
PIPO	Phase In/Phase Out
PM	Program Manager
POC	Point of Contact
PRS	Performance Requirements Summary
PWS	Performance Work Statement
QA	Quality Assurance
QAP	Quality Assurance Program

QC	Quality Control
QCP	Quality Control Plan
RGB	Red, Green, Blue
SAR	Synthetic Aperture Radar
SCI	Sensitive Compartmented Information
SIGINT	Signals Intelligence
SOCOM	U.S. Special Operations Command
SOP	Standard Operating Procedure
SOUTHCOM	U.S. Southern Command
SOW	Statement of work
TBO	Time Before Overhaul
TE	Technical Exhibit
TM	Technical Monitor
TOT	Time over Target
TPOC	Technical Point of Contact
TS	Top Secret
U.S.	United States
USACE	United States Army Corp of Engineers
WBS	Work Breakdown Structure

Part 3. Government Furnished Information, Equipment, Services and Facilities

3.1.1 General

3.1.2 The Government will provide CAC cards for Contractor personnel.

3.1.3 The Government will provide Synchronized Pre-deployment Operational Tracker (SPON) coordination for any OCONUS travel.

3.2 The Government will provide the following information:

3.2.1 DD 1423 incorporated as contract attachment.

3.2.3 DoD flight information publications, updates, and supplements appropriate to the area of operations thirty (30) days after contract award.

3.2.4 A Letter of Authorization (LOA) for Contractor personnel to permit attendance at a CONUS Replacement Center (CRC) or similar approved replacement center, in accordance with DoD guidance. The LOA will validate, at a minimum, their Contractor employment, GS equivalency (GS 12 or 13), and authority to process within twenty (20) days of contract award.

3.2.5 Applicable installation, facility, and area commander installation and facility access and local security policies and procedures upon deployment date.

3.2.6 Requirements for AOR-specific AT awareness training and local deployment requirements and documentation. The COR will provide this information (required training, country clearance requirements, and local deployment requirements and documentation) as GFI at contract award.

3.2.7 Local AT/OPSEC policies, including required compliance with laws and regulations, pre-deployment requirements, required training (per combatant command guidance), and personnel data requirements provided by the COR as GFI at contract award.

3.2.8 Country and theater clearance requirements and combatant commander and subordinate task force commander policies and directives provided by the COR as GFI at contract award.

3.2.10 Specific requirements that must be met for access to Government information systems as GFI at contract award.

3.2.11 CCMD documentation with approval to deploy and operate the 3DH2-M mission in support of the CCMD, for example Operation Orders (OPORDS), Fragmentary Orders (FRAGOS), and Concept of Operations (CONOPS).

3.3 The Government will provide all required sensors and cameras, including but not limited to, the list below:

Sensor	Total Units
• Optech Galaxy T2000	13 or more
• Leica City Mapper 2	One or more
• Leica ALS80UP	11 or more

• Leica SPL 100HA	Two or more
• Leica SPL 100	Two or more
• Chiroptera-5 LiDAR	One or more
• Sigma Space High Altitude Speed, and Resolution Photonic (HARP) LiDAR	One or more
• Phase One iXM EO Cameras	35 or more
• Phase One SPARC	Two or more
• Vision Map Mist	One or more
• MX-25	One or more
• 3DEO Sequoia LiDAR	Two or more
• Leonardo Osprey 30 AESA Radar	One or more
• Parson ICS 550	One or more
• DRT 3300	One or more
• Lanshark/Sharkcage System	One or more
• ViaSAT Communications	One or more
• Starlink Airborne System	Two or more
• L3 Harris Wescam MX25	One or more
• MDRMN	One or more
• Mayflower Magna F system	7 or more
• Controller CC3X	One or more
• Operator Console OC60	One or more
• Laser Controller 1U Crystal RS112S14	One or more
• TermoTek Chiller	One or more
• GSM4000	One or more
• Crystal 2U LIDAR OBP Computer	One or more
• Crystal 2U EO OBP Computer	One or more
• KIV-77	One or more
• TA-24 SAASM GPS Receiver	One or more
• GPS AT1675-180	One or more
• VR-18	One or more
• VORTEX TC DL	One or more
• TNT	One or more
• HTL	One or more
• Delta Labs Processor (DLP)	One or more
• DAEDALUS	One or more
• INGRES	One or more
• Telephonies RDR-1700G(V)2	One or more
• Starlink Aero Terminal	One or more

These items will be provided to the contractor at the deployment sites no later than 7 days prior to the start of the base period.

3.3.1 All Government Off the Shelf (GOTS) and Commercial Off the Shelf (COTS) software that is required for use in operation of the system. It has been established that this software is compatible with Government-approved networks and flight systems. GOTS/COTS software items are listed in Table 1:

Table 1

Developer	Software Title	Use
Applanix	POSAV	Data Processing
Applanix	POSPacMMS	Data Processing
Applied Imagery	QTModeler	Data Processing and Exploitation
BAE	SOCETGXP	Data Exploitation
Bentley	Connection Client	Data Processing
Bentley	Microstation	Data Processing
Blue Marble Geographies	Global Mapper Pro	Data Management and Exploitation
Dassault Systems	Solidworks 2024	Data Processing
ESRI	ArcGIS Desktop	Mission Planning, Data Management, and Product Generation
ESRI	ArcGIS Pro	Mission Planning, Data Management, and Product Generation
Hexagon	HxMap	Data Processing
Hexagon	Lidar Survey Studio (LSS)	Data Processing
Hexagon	Lidar Survey Studio Lie Server	Data Processing
Inpho	Orthovita - Trimble Photogrammetry Suite	Data Processing
Leica	CLM (Leica License Manager)	Data Processing
Leica	MissionPro	Data Processing
LizardTech	GeoExpress	Data Management
Microsoft	Microsoft Office	Program Management
Novatel	Inertial Explorer	Data Processing
OGSystems	TwoEye	Data Processing
PhaseOne	CaptureOne	Data Processing
PhaseOne	CaptureOne Pro	Data Processing
PhaseOne	iX Plan	Data Processing
Simactive	Correlator3d	Data Processing
Skyline	Photomesh	Data Processing
Skyline	TerraExplorer Pro	Data Processing
Teledyne Optech	AirborneMissionManager	Data Processing

Teledyne Optech	Diskextract	Data Processing
Teledyne Optech	FMS-NAV	Included with Hardware
Teledyne Optech	LMS	Data Processing
TerraSolid	Spatix	Data Processing
TerraSolid	TerraMatch	Data Processing
TerraSolid	TerraModeler	Data Processing
TerraSolid	TerraPhoto	Data Processing
TerraSolid	TerraScan	Data Processing
TerraSolid	TerraSolid License Manager	Data Processing

3.3.2 The government will provide the following communication systems and radios at deployment sites NLT 7 days prior to the start of the base period.

- Harris PRC 117G (14 or more)
- CSEL radios (18 or more)
- SKL (5 or more)
- APX-119 (2 or more)

3.3.3 The government will provide Airborne Laser Terrain Mapping (ALTM) software and any other International Traffic in Arms Regulations (ITAR) -restricted equipment or information.

3.4 The Government Combatant Command (COCOM) will provide the following services and facilities:

3.4.1 Mission scheduling and planning sufficient to allow Air Operations Center airspace coordination and ATO development within 72 hours prior to mission.

3.4.2 Coordination with Tactical Air Control facilities for emergency mission requirements within twelve (12) hours of mission.

3.4.3 Access to Coalition Forces (CF) weather facilities daily prior to mission.

3.4.4 Access to mission profile intelligence briefing for threat identification and assessment daily prior to mission.

3.4.5 Access to CF airfields as required for mission performance daily prior to mission.

3.4.6 The Government Contracting Officer will appoint a GFR to monitor the Contractor's flight and ground operations. At its unilateral discretion, the Government may incorporate DFARS 252.228.7001 Ground and Flight Risk.

3.4.7 Flight oversight of all airspace and aviation procedures by the OCONUS Technical Manager (TM) and Contract Officer Appointed GFR or designated equivalent daily prior to mission.

3.4.8 Flight profiles that will avoid areas where hostile air threat activities are known to exist or are anticipated to occur based on current threat assessments provided daily prior to mission.

3.4.9 A secure base of operations daily from start to finish of deployment.

3.4.10 The Government will provide all necessary utilities (e.g., water, power, fuel) to support operations in areas where they are not commercially available to the Contractor daily from start to finish of deployment.

3.4.11 Housing and office locations at forward deployed locations while operating in areas where these items are not commercially available daily from start to finish of deployment. In addition, the Government will be required to provide housing if the Operational Commander requires the Contractor personnel to live on a base or FOB due to operational/Force Protection reasons or other mission requirements.

3.4.12 A secure OCONUS facility for any data processing of classified data daily from start to finish of deployment.

3.4.13 Adequate office space that includes adequate power supply and cooling for Contractor personnel who are required to work onsite at Government facilities as part of performing this effort daily from start to finish of deployment.

3.4.14 The Government will provide security for personnel, and aircraft when operating in any environment such as an active combat zone where continuing contingency operations are ongoing daily from start to finish of deployment.

3.4.15 A group Contractor Replacement Center (CRC) class (or equivalent) to begin no later than Day 30 of the transition period.

3.4.16 OCONUS HAZMAT and shipping inspectors as required for remobilization thirty (30) days from final date of remobilization.

3.4.17 Military transport as required for mobilization, remobilization, and shipment of spares and equipment depending upon the theater of operations as required.

3.5 The Government Combatant Command (COCOM) may provide the following services and facilities:

3.4.1 SI and/or multi INT sensor operators.

3.4.2 Onboard data analysts.

Part 4. Contractor Furnished Items and Responsibilities

4.1 General

4.1.1 The Contractor shall furnish everything required to perform the requirements of this PWS except for the items listed that will be provided by the Government in Part 3 of this PWS.

4.2 The Contractor shall furnish the following equipment: Two (2) turboprop and two (2) jet Federal Aviation Administration (FAA)--certified aircraft that meet the 3DH2-M operational requirements and support equipment to maintain the safe operation of the aircraft. The Contractor shall assume total responsibility for the maintenance and repair of the multi-mission aircraft.

4.2.1 The collection operations will be comprised of the following fleet:
Two (2) manned (turboprop) airborne 3DH2-M BE I Light Detection and Ranging (LiDAR)/Electro-Optical (EO) collection systems.
Two (2) manned (Jet) airborne 3DH2-M BE III collections systems.

4.2.2 The Contractor shall provide and maintain manned fixed wing platforms to include the following:

4.2.3 The Contractor shall provide two (2) FAA-certified aircraft, that must fulfill all Army AWR requirements in compliance with DCMA 8210, fully modified and equipped to provide services as an airborne mapping platform and capable of continuously travelling 750 NM from its base of operations to its collection area, performing a one-hour on-station collection, and returning to base. These aircraft will be used for the sensor and camera configurations to meet the PWS requirements for the BuckEye I (BE I) systems.

4.2.4 The Contractor shall provide two (2) FAA-certified aircraft, that must fulfill all Army AWR requirements in compliance with DCMA 8210, fully modified and equipped to provide services as an airborne mapping platform capable of continuously travelling 3,000 NM from its base of operations to its collection area, performing a four-hour on-station collection, and returning to base. This aircraft will only be used for the sensor and camera configurations to meet the PWS requirements for the BE III system.

4.2.4.1 Any aircraft supporting a BE III system as well as B kit architecture and integration shall meet all accreditation standards to operate as a tactical SCIF (TSCIF).

4.2.4.2 The contractor shall operate BE III aircraft as a TSCIF as directed by the Government.

4.2.4.3 The contractor shall integrate sensors, either from section 3.3 of this PWS or as identified by the Government, into Unmanned Aircraft identified by the Government.

4.3 The Contractor shall develop and document a sparing strategy for manned and unmanned platforms. This sparing strategy will include recommendations for Line Replaceable Units (LRUs) at the bases of operations, and identification of long lead-time components and subassemblies to be procured and held at the Contractor's CONUS facility or other suitable location, CONUS or OCONUS.

4.3.1 The Contractor shall procure spare parts for the platforms in accordance with the sparing strategies.

4.3.2 The Contractor shall provide the personnel and equipment required to maintain the platforms.

4.3.3 The Contractor shall assume total responsibility for the maintenance and repair of the platforms, and all other Contractor Furnished Equipment (CFE) required to perform the required tasks.

4.4 The Contractor will operate aircraft under a State Use Designation with the US Army serving as the Air Worthiness Release (AWR) Authority.

4.4.1 The contractor shall be bound by the operating procedures contained in the combined regulation/instruction entitled "Contractor's Flight and Ground Operations" DCMA INST 8210-1D, AFI 10-220, AR 95-20, NAVAIRINST 3710.1H, COMDTINST M13020.3B, February 6, 2023 for manned and unmanned aircraft operations, in effect on the date of award, and the applicable updated regulation/instruction during performance as stated in the PWS.

4.4.2 The Contractor shall not bill the Government for delays caused solely by the Contractor's internal reviews, including but not limited to post-modification aircraft safety of flight reviews.

4.4.3 The Approving Authority will appoint a GFR to monitor the contractor's flight and ground operations in accordance with DCMA INST 8210-1D, AFI 10-220, AR 95-20, NAVAIRINST 3710.1H, COMDTINST M13020.3B, February 6, 2023 and DFARS 252.228-7007 Public Aircraft and State Aircraft Operations-Liability.

4.4.4 Public Aircraft Operations/State Liability - The Contractor shall assume all liability, maintain adequate insurance, Workers' Compensation and not hold the Government liable in the event of aircraft damage or injury to personnel IAW DFARS Clause 252.228-7007.

Part 5. Specific Tasks

5.1 General

The Contractor shall provide all labor, equipment, technical, managerial, logistical, and engineering support and services to support the U.S. Government in the execution of its 3DH2-M operation and sustainment mission. The Contractor shall provide personnel possessing the skills, knowledge, certifications, and training to satisfactorily perform the services required by this PWS. The Contractor shall procure and manage all materials, supplies, and services needed to perform the tasks specified in this PWS. The remainder of this section describes the specific tasks and activities that the Contractor shall conduct. All acronyms are spelled out as follows and are listed as solely the acronym thereafter.

5.2 Technical and Operations Tasks

5.2.1 Locations (OCONUS and CONUS)

BEI

Australia (1 MFW - BE I) in IndoPACOM AOR

South Korea (1 MFW – BE I) in IndoPACOM AOR

BE III

Camp Lemonnier (1 MFW - BE III System) in AFRICOM AOR

Each OCONUS base camp/deployed site in the list above is within a CCMD AOR. CCMD AORs and their geographic limitations are detailed further in Figure 1 of the PWS.

The geographic locations of data collection missions are defined in the CCMD generated OPORD/TASKORD/EXORDs received by the Government. Operations within each AOR may require conducting operations temporarily from a 'spoke location' as directed by the Government.



Figure 1

5.2.1.2 CONUS Test and Evaluation (T&E) Locations are listed below: HR3D MOA, southern Ohio Valentine MOA, Marfa, Texas Tombstone MOA, Arizona/Texas border.

5.2.1.3 CONUS Basing Location:
TBD (1 MFW - BE I when available) in NORTHCOM AOR

5.2.1.4 One (1) MFW when available may be used for CONUS-based operations at the levels of effort stated in PWS Section 5.5.5

5.2.2 Manned Fixed Wing Platforms Collection and Processing

The Contractor shall provide technical, management, engineering, maintenance, testing, training, and operational support that include the following:

5.2.2.1 Collection

5.2.2.1.1 The Contractor shall operate the GFP equipment and software identified in Part 3 of this PWS.

5.2.2.1.2 The Contractor shall plan and conduct 3DH2-M data collection missions upon receipt from CCMD Collection Management (CM)/Air Operations directives.

5.2.2.1.3 The Contractor shall keep and make available for Government inspection, a logbook record of all flights performed.

5.2.2.1.4 The Contractor will provide the ability to collect airborne data using contractor owned aircraft capable of integrating Government furnished sensors and capable of collecting a minimum of 300 sq/km per flight/aircraft for BE I; 2000 sq/km per flight/aircraft for BEIII within 120 days of contract award.

5.2.2.1.5 The Contractor shall maintain a 90% Operational Readiness rate to respond to missions from the Government within 24 hours.

5.2.2.1.6 The Contractor shall perform collection missions in accordance with the OPORDS for each of the performance locations identified in paragraph 5.2.1. Collection missions will be directed by the Government and may take place during daylight or nighttime hours. Daylight collection mission flight hours will optimize sun angles for the highest quality EO imagery. The priority of data collection is in accordance with the CCMD data requests.

5.2.2.1.7 A flight or a series of flights (also known as sortie or sorties), with a fully mission capable (FMC) maintained aircraft (having departed from a deployed location or base camp) flying at approved altitudes, with flight crew on board, sensors and cameras fully mission capable/maintained and operational, transiting to a location, collecting data over one or more location/area provided by the CCMD CM, transiting from area of collection to return to the deployed location/base camp. Each mission flight time total is subject to and in accordance with the requirements in DFARS 252.228- 7007, IAW DCMA 8210 and AR 95-1.

5.2.2.1.8 The Contractor shall accommodate a standard operational flight profile of 8,000 feet (ft) AGL to 19,000 ft AGL for the BE I systems; and 15,000 ft AGL to 35,000 ft AGL for the BE III systems and accommodate mutually agreeable flight profiles outside of these bands based on operational, sensor, and aircraft limitations.

5.2.2.1.9 The Contractor shall collect 1 m LIDAR and 10 cm EO imagery or better with the manned fixed wing systems and 50cm LIDAR and 5cm EO imagery or better with unmanned systems

5.2.2.2 Geospatial Data Processing

The Contractor shall provide geospatial data processing, analysis, exploitation and meshing of various phenomenology to create and deliver 2, 2.5 and 3D content for use by Army Programs of Record.

Provide large scale worldwide advanced cloud based geospatial processing and dissemination in a rapid fashion, between 24-48 hours.

5.2.2.2.1 LIDAR Processing

The Contractor shall provide LIDAR processing to include the following:

5.2.2.2.1.2 The Contractor shall process airborne LIDAR following established guidelines developed by AGC.

5.2.2.2.1.3 The Contractor shall, after raw sensor data and any required GPS correction data is available at the data processing facility, process the LiDAR data for each mission in less than 24

hours on average, given factors of mission size (km²), point density (points per meter), and system characteristics (BEI and BEIII). For missions with co-collected EO data, the Contractor shall make available to the Government non-colored LiDAR data within the allotted processing time, and colored-LiDAR data within 12 hours after EO processing is complete.

5.2.2.2.1.4 The Contractor shall maintain sufficient hardware at CONUS and OCONUS locations to meet production demands, perform quality control, and maintain lab hardware and software.

5.2.2.2.1.5 The Contractor shall provide LiDAR final products as LAZ 1.4 datasets in Cloud Optimized Point Cloud (COPC) format. The Contractor shall colorize the point clouds (r,g,b), for missions with co-collected imagery.

5.2.2.2.1.6 The Contractor shall process the LiDAR Point Cloud data into Digital Surface Models (DSMs) and Digital Terrain Models (DTMs), at resolutions approximating the average point cloud spacing data (e.g., manned airborne collected point cloud data at 1-meter average point spacing will be gridded at 1-meter). The DSM and DTM data shall be delivered as 32-bit floating point format.

5.2.2.2.1.7 For LIDAR sensors that support intensity images, the Contractor shall provide LIDAR intensity images depicting LIDAR reflectivity of the surface topography in grayscale as derived from the LIDAR point cloud tiles. The Contractor shall deliver intensity images as 8-bit grayscale format.

5.2.2.2.1.8 The Contractor shall deliver the DSM, DTM, and Intensity data in Cloud Optimized GeoTIFF (COG) with LZW Compression (or alternate as defined by the Government).

5.2.2.2.1.9 The Contractor shall deliver all elevation data (Point Clouds, DSM, and DTM) in the EGM08 vertical datum.

5.2.2.2.10 The Contractor shall ensure all point cloud LiDAR Products point cloud data are tiled to a 1km x 1km or alternate tile schema as defined by the Government.

5.2.2.2.11 The Contractor shall ensure all raster LiDAR Products (DSM, DTM, and Intensity Image) are tiled to a 5km x 5km or alternate tile schema as defined by the Government.

5.2.2.2.12 The Contractor shall ensure all raster LiDAR Products (DSM, DTM, and Intensity Image) have a 10-meter overlap with adjacent tiles, including tiles yet to be collected.

5.2.2.2.13 The Contractor shall provide a copy of all raw LiDAR and associated collection data for each mission.

5.2.2.2.2 The Contractor shall provide Electro Optical (EO) processing to include the following:

5.2.2.2.2.1 The Contractor shall process all manned aircraft collected EO imagery following established guidelines developed by AGC.

5.2.2.2.2.2 The Contractor shall process the EO data in less than 72 hours on average for the BE I, BE II, BE III systems using GOTS and COTS GFP software.

5.2.2.2.2.3 The Contractor shall maintain sufficient hardware at CONUS and OCONUS locations to meet production demands, perform quality control, and maintain lab hardware and software.

5.2.2.2.2.4 The Contractor shall have a capability to process EO data using computer equipment and personnel located at the flight operations base.

5.2.2.2.2.5 The Contractor shall provide a copy of all raw EO images and associated collection data for each mission.

5.2.2.2.2.6 The Contractor shall provide, upon request by the Government, a copy of all intermediary developed individual camera frames; color-corrected, in uncompressed TIFF Format, and with calculated position and orientation information.

5.2.2.2.2.7 The Contractor shall provide orthomosaic imagery products as color-balanced, georegistered, orthorectified wide area image mosaics in uncompressed GeoTIFF and Cloud Optimized GeoTIFF (COG) formats.

5.2.2.2.2.8 The Contractor shall, for missions with co-collected LiDAR and EO, tile orthomosaic imagery products to a 5km x 5km tiling schema, identical to the LiDAR tiling schema, or as alternatively defined by the Government.

5.2.2.2.2.9 The Contractor shall ensure all orthomosaic imagery products have a 10-meter overlap with adjacent tiles, including tiles yet to be collected.

Annual COTS software maintenance subscriptions shall include the following, at a minimum listed in Table 2:

Table 2

Developer	Software Title	Quantity
Applanix	POSAV	Unlimited
Applanix	POSPacMMS	28
Applied Imagery	QTModeler	89
BAE	SOCETGXP	16
Bentley	Connection Client	Unlimited
Bentley	Microstation	54
Blue Marble Geographies	Global Mapper Pro	30
Dassault Systems	Solidworks 2024	2
ESRI	ArcGIS Desktop w/ 3D & Spatial Analyst Extensions	15
ESRI	ArcGIS Pro w/ 3D & Spatial Analyst Extensions	5

Hexagon	HxMap	6
Hexagon	Lidar Survey Studio (LSS)	Unlimited
Hexagon	Lidar Survey Studio Lie Server	Unlimited
Inpho	Orthovita - Trimble Photogrammetry Suite	80
Leica	CLM (Leica License Manager)	Unlimited
Leica	MissionPro	23
LizardTech	GeoExpress	5
Microsoft	Microsoft Office	Unlimited
Novatel	Inertial Explorer	45
OGSystems	TwoEye	Unlimited
PhaseOne	CaptureOne	Unlimited
PhaseOne	CaptureOne Pro	10
PhaseOne	iX Plan	7
Simactive	Correlator3d	45
Skyline	Photomesh	2
Skyline	TerraExplorer Pro	4
Teledyne Optech	AirborneMissionManager	9
Teledyne Optech	Diskextract	Unlimited
Teledyne Optech	FMS-NAV	8
Teledyne Optech	LMS	24
TerraSolid	Spatix	68
TerraSolid	TerraMatch	53
TerraSolid	TerraModeler	2
TerraSolid	TerraPhoto	2
TerraSolid	TerraScan	53
TerraSolid	TerraSolid License Manager	Unlimited

5.2.2.2.3 The Contractor shall develop and document a sparing strategy for the platforms.

5.2.2.2.3.1 This sparing strategy will include recommendations for LRUs at the bases of operations, and identification of long lead-time components and subassemblies to be procured and held at the Contractor's CONUS facility or other suitable location, CONUS or OCONUS.

5.2.2.2.3.2 The Contractor shall procure spare parts for the platforms in accordance with the sparing strategies.

5.2.2.2.4 The Contractor shall provide the personnel and equipment required to maintain the platforms.

5.2.2.2.5 The Contractor shall assume total responsibility for the maintenance and repair of the platforms and all other CFE required to perform the required tasks.

5.2.2.2.6 The Contractor shall integrate the GFP sensors to the aircraft platforms.

5.2.3 The contractor shall provide data exploitation as directed by the Government

5.2.4 The Contractor shall provide OCONUS Manning

5.2.4.1 The manning footprint for each OCONUS basing location shall be approved by the Government and is subject to change as directed by the Government to meet CCMD needs:

5.2.4.2 The contractor shall limit flight crew labor hours to eight (8) hours on all crew rest days and/or days an individual is not scheduled to fly.

5.2.4.3 The contractor shall operate jet aircraft up to a maximum of 2,400 flight hours and turboprop aircraft up to a maximum of 1,800 flight hours in a 365 day period or as directed by the Government

5.2.5 The Contractor shall maintain all sensors and systems installed in or operated onboard the aircraft as follows:

5.2.5.1 The Contractor shall check and maintain a current calibration for installed and/or operated GFP sensors. The Contractor shall perform the calibration adjustments upon installation of the GFP sensors into the aircraft and then monthly thereafter.

5.2.5.2 The Contractor shall procure annual warranty or maintenance plans for each of the GFP sensors.

5.2.5.3 The Contractor shall monitor GFP sensor health and data quality.

5.2.5.4 The Contractor shall follow the manufacturer recommended maintenance schedule and operations manuals for all GFP sensors.

5.2.5.5 The Contractor shall develop and document a sparing strategy for the GFP sensors. This sparing strategy will include recommendations for LRUs at the bases of operations, and identification of long lead-time components and subassemblies to be procured and held at the Contractor's CONUS facility or other suitable location, CONUS or OCONUS.

5.2.5.6 The Contractor shall procure spare parts for the GFP sensors in accordance with the sparing strategies. The contractor shall take all actions necessary to adjust for overages, shortages, damage and/or other discrepancies discovered upon receipt, in shipment of Contractor Acquired Property (CAP) from a vendor or supplier, so as to ensure the proper allocability and allowability of associated costs. The spare parts will be retained by the contractor for use after inspection and acceptance by the Government and shall be made accountable to the contract through a contract modification listing the item(s) as Government-furnished property. The contractor shall maintain accountability of all CAP the same as GFP until the Government takes title. All purchase orders must be approved by the Government prior to ordering.

5.2.5.7

5.2.6 Dissemination, Storage, and Maintenance

5.2.6.1 The Contractor shall deliver all processed data on hard drives, to the designated Government CONUS location and representative.

5.2.6.2 The Contractor shall provide personnel for the CONUS/OCONUS dissemination of manned airborne LIDAR/EO products as required by the Government. The Contractor shall communicate with local NGA and other Government representatives to organize the rapid delivery into their established dissemination systems.

5.2.7 CONUS Based Data Content Management

The Contractor shall perform the following tasks for CONUS Based Data Content Management:

5.2.7.1 Provide strategic and operational level technical guidance.

5.2.7.2 Produce data management plans, strategies to enable enterprise level sharing.

5.2.7.3 Develop and provide metrics to evaluate data utilization and compliance.

5.2.7.4 Assess data processes and procedures for their effectiveness, efficiency, and consistency as applied to the geospatial area.

5.2.7.5 Define, publish, and maintain official Army geospatial data metrics, taxonomies, semantics, ontologies, metadata standards, analytical processes, and transformation functions.

5.2.7.6 Define, publish, and maintain common Army geospatial data management procedures for geospatial data and system administration.

5.2.7.7 Define, publish, and maintain common Army geospatial data management procedures for geospatial operations, including planning and direction, collection, processing and exploitation,

analysis and production, dissemination and integration, utilization, evaluation, and feedback.

5.2.7.8 Identify, maintain, and disseminate Army geospatial data extraction procedures and content specifications for non-standard entity data.

5.2.7.9 Identify, maintain, and disseminate Army geospatial data operational issues and lessons learned.

5.2.7.10 Identify and recommend Army geospatial authoritative data sources to the Geospatial Data Steward.

5.2.7.11 Provide configuration management of Standard Shareable Geospatial Foundations (SSGF) data technical guidance artifacts for materiel developers.

5.2.7.12 Be responsible for the quality of geospatial information presented in mission orders.

5.2.7.13 Identify gaps in geospatial data.

5.2.7.14 Be responsible for establishing and standardizing the procedures for populating the database within the AGE.

5.2.8 Authority to Operate

5.2.8.1 The contractor shall support Cybersecurity Assessment and Authorization (A&A) and Connectivity or Interconnectivity activities as required to obtain ATOs. This applies to systems that have not been successfully assessed and authorized and to systems that undergo modifications that impact the Cybersecurity and ATO.

5.2.8.2 The contractor shall follow DODI 8510.01 "Risk Management Framework (RMF) for DoD Information Technology (IT)", NIST 800-37 "Guide for Applying the Risk Management Framework to Federal Information Systems", and Intelligence Community Directive (ICD) 503, "Intelligence Community Information Technology Systems Security Risk Management". The Contractor shall implement IA security controls to safeguard the system, data and software IAW Certification and Accreditation, the Committee on National Security Systems (CNSS) Instruction No. 1253, "Security Categorization and Control Selection for National Security Systems" and NIST 800-53, "Recommended Security Controls for Federal Information Systems and Organizations" as well as any applicable overlays that are determined to be applicable to the system. The authorizing agency/authority will be determined via coordination with the Government.

5.2.8.3 The contractor shall generate and provide all necessary data and documentation to obtain all necessary ATOs.

5.2.8.4 The contractor shall perform all required activities to maintain system accreditations in accordance with Federal Information Security Modernization Act (FISMA).

5.2.8.5 The Contractor shall start system reauthorization process one hundred eighty (180) calendar days prior to the current ATO expiration. The Contractor must obtain updated ATO to ensure continued operation as required.

5.2.8.6 The Contractor shall deploy regularly scheduled and ad hoc patches to field sites and System Integration Laboratories (SILs). The Contractor shall be required to maintain system access and user accounts needed to install patches and perform system scans. The Contractor

shall coordinate with the Government and field site personnel (both Contractor and Army) prior to deploying any patches or deploying personnel to assist. When installing patches, the Contractor shall keep records of all systems and hard drives patched, recording any failures to patch, to include all primary, backup, and test hard drives, and provide this information to the Government after each patch is deployed. The patch status of all systems and all hard drives (including SILs) shall be maintained in a database by the Contractor and made available to the Government upon request. The Contractor shall update all documentation as required after each patch, including but not limited to SSPs and SVDs and provide updated documentation to the Government.

5.2.8.7 The Contractor shall report all IAVAs, STIGs, and Bulletins IAW the Vulnerability Management Plan. The Contractor shall provide a list outlining which IAVAs were implemented, those not implemented using an IAVA tracking spreadsheet. All non-implemented IAVAs and STIGs shall have Government concurrence. The Contractor shall provide a comprehensive and up to date security scan using approved scanning tool (e.g. Nessus) IAW system Vulnerability Management Plan.

5.3 Mobilization & Remobilization

5.3.1 Mobilization

5.3.1.1 Mobilization is the process of preparing platforms, sensors, processing hardware and software, associated equipment, and personnel to mission ready status at a base camp/deployed location of operations.

5.3.1.2 Pre-Deployment/Pre-Mobilization CCMD Collection Deck

Prior to mobilization by CCMD staff personnel, data collection requests within the CCMD are compiled within established CCMD protocols and provided to the Government. The Government will provide to the Contractor the desired CCMD Collection Deck. Upon receipt of the desired CCMD Collection Deck, the Contractor will provide the following to the Government, within ten business days from receipt:

- Estimated number of collection bounding boxes
- Estimated number of flight lines within each bounding box
- Estimated hours to complete data collection for each bounding box
- On Location/Deployment Operations Collection Deck Schedule

Upon completion of the data collection plan estimates and Government approval, the Contractor Lead at each deployed location will begin implementation of collection plans in accordance with CCMD flight operations and regulations. After initial implementation of the CCMD Collection Deck, each deployed location Contractor Lead will be provided updated Collection Decks weekly, by the CCMD Collection Managers through established protocols and software applications. The CCMD will identify the collection decks' priority and whether they are time critical, ad hoc operational and/or non- operational. The Contractor will perform collection planning as stated above throughout the period of performance.

5.3.1.3 The Contractor shall communicate with the Government to organize, schedule, and receive CRC or equivalent training.

5.3.1.4 The Contractor shall place orders for all materials required for initial missions and expedite long lead material items.

5.3.1.5 The Contractor shall conduct PDSSs for all sites and identify any available GSE to potentially receive aircraft early.

5.3.1.6 The Contractor shall pack and ship by commercial air transportation all pallets and CONEX containers for shipment containing spares, GSE, and tools.

5.3.1.7 The Contractor shall receive all procured items at its CONUS/OCONUS facilities/basing sites.

5.3.1.8 The Contractor shall ship boxes and pallets containing all data processing equipment, aircraft spare parts, sensor spare parts, and all other required equipment and materials.

5.3.1.9 The Contractor deploying team shall receive CRC or equivalent training.

5.3.1.11 The Contractor shall ensure all GSE is organized and ready for aircraft arrival.

5.3.1.12 The Contractor shall receive all data processing equipment, aircraft spare parts, sensor spare parts, and all other required equipment and materials at the OCONUS sites.

5.3.1.13 The Contractor shall ferry the fixed wing aircraft to theater.

5.3.1.14 The Contractor's first rotation deploying team shall arrive at the OCONUS sites with adequate time to begin operational missions on the required date.

5.3.1.15 The Contractor shall receive the aircraft at the OCONUS sites.

5.3.1.16 The Contractor shall prepare the OCONUS sites to full operational state.

5.3.1.17 The Contractor shall ensure that all personnel have the appropriate in-theater orientation and training.

5.3.1.18 The Contractor shall ensure the flight operations personnel hold the necessary flight qualifications.

5.3.1.19 The Contractor shall provide support to the host command to ensure the aircraft are released for flight operations.

5.3.1.20 The Contractor shall begin the 3DH2-M data collection and processing at the OCONUS sites no later than the conclusion of the transition phase, if required.

5.3.2 Remobilization

Remobilization is the process of moving the platforms, sensors, processing hardware and software, associated equipment, and personnel from one location to another and preparing them to mission ready status.

5.3.2.1 Phase I- Redeployment of the aircraft and onboard sensors and mission equipment as follows:

5.3.2.1.1 The Contractor shall develop a detailed remobilization plan and deliver it to the Government PM and COR one (1) week after Government notification to the Contractor of a remobilization.

5.3.2.1.2 The Contractor shall update the remobilization plan at least weekly during remobilization.

5.3.2.1.3 The Contractor shall de-install the GFP sensors and other equipment as required and prepare the aircraft for ferry. The planning and processing for shipments of GFP that are being redeployed shall be outlined in the Contractor's Property Control Plan.

5.3.2.1.4 The Contractor shall arrange the flight plan and diplomatic clearances.

5.3.2.1.5 The Contractor shall provide pilots and ferry the aircraft to the destination site.

5.3.2.1.6 The Contractor shall complete Phase I within 14 days of remobilization initiation or as required.

5.3.2.2 Phase II-Inventory, packing, and arranging transportation to include the following:

5.3.2.2.1 The Contractor shall initiate an inventory of all equipment and materials to be remobilized and provide copies to the KO, GPA, and COR.

5.3.2.2.2 The Contractor shall initiate, for disposition instructions any Government-furnished property that is excess and/or will not be required to return to CONUS or to be transferred to another entity in theatre, through the GFP Module within PEE to the Plant Clearance Officer. Approval is required for turn-in or transfer of any GFP.

5.3.2.2.3 The Contractor shall load the equipment and materials onto pallets and into CONEXs.

5.3.2.2.4 The Contractor shall organize transportation and ensure that all customs procedures are met for shipping the equipment.

5.3.2.2.5 The Contractor shall redeploy personnel to the destination site.

5.3.2.2.6 The Contractor shall complete Phase II within thirty (30) days of remobilization initiation or as required.

5.3.2.3 Phase III-Inventory, HAZMAT, and transportation: This phase is completely dependent on the availability of Government HAZMAT and shipping inspectors and on military and commercial transportation. Planning time for completion is the thirty (30)- day to sixty (60)- day mark.

5.3.2.3.1 The Contractor shall complete the inventory initiated in Phase II.

5.3.2.3.2 The Contractor shall organize HAZMAT inspections.

5.3.2.3.3 The Contractor shall ensure that all sensitive equipment and information are properly inventoried, packaged, and shipped by an approved means. The inventory shall be provided

to the KO, GPA, and COR prior to packaging and shipping. The contractor will have a plan in place for the accountability and shipment of any sensitive items outlined within the contractor's property control plan.

5.3.2.3.4 The Contractor shall ship CONEXs and pallets.

5.3.2.3.5 The Contractor shall redeploy personnel to the destination site.

5.4 Transition

The Contractor shall ensure operational capability NLT one hundred twenty (120) days post award, in the required theater and countries, with associated GFE/GFP, approvals/accreditation, and necessary deployed and CONUS personnel.

5.5 CONUS Training and Support

5.5.1 The Contractor shall establish and maintain the capability to train qualified personnel for any deployed collection, processing, or leadership position.

5.5.2 The Contractor shall train qualified personnel for any deployed position as required.

5.5.3 The Contractor shall provide other CONUS support to the program, such as data collection to refine the end products, quality assurance of delivered products, rework of deficient products, surge data processing when the operational collection outpaces the site-based processing, and refinement of delivered products by HITL value-added processing (e.g., LIDAR bare earth extraction, EO radiometric matching).

5.5.4 The Contractor shall conduct sensor test and evaluation at the CONUS Contractor facility, Military Operations Area (MOAs) and the military installations listed in paragraph 5.2.1.2 and 5.2.1.3.

5.5.5 The Contractor, for the purpose of CONUS test and evaluation and military installation data collection, shall provide the personnel and equipment required to provide the Government with a maximum of 25 flight hours per month per aircraft.

5.6 Program Management Tasks

5.6.1 Program Management Reporting

5.6.1.1 The Contractor shall provide accurate and timely schedule and performance information throughout the life of the program.

5.6.1.2 The Contractor shall provide a CWBS and update if necessary.

5.6.1.3 The Contractor shall provide an IMS and update it monthly.

5.6.1.4 The Contractor shall ensure the AGC PM Office is copied on emails, briefing notes, and all Contractor distributed materials. Additionally, the Contractor will copy the AGC PM Office on all Contractor engagement correspondence with CCMD personnel, customers, Requirements and Collection Managers.

5.6.2 Risk Management

The Contractor shall establish a risk management system through the integration of metrics to monitor program status to mitigate program risks and provide for special emphasis on software development efforts.

5.6.3 Configuration Management

The Contractor shall establish a comprehensive configuration management system, in conjunction with the Government/Program Office, for equipment, spares, repairs, COTS and GOTS software versions, schedule, WBS, IMS, risk register, raw sensed data, and processed data products.

5.6.4 Key Personnel

The Contractors shall provide five Key Personnel: Program Manager, Chief Engineer, Chief Pilot, Chief Mechanic, and Chief Operations Officer. **NOTE:** The Key Personnel identified must be used on the project. No deviations will be permitted unless they are expressly approved by the Contracting Officer. An offeror's use of different personnel than those proposed without approval from the Contracting Officer will be grounds for a Termination for Default.

5.6.4.1 Program Manager

The PM is responsible for the performance of the requirements of this PWS. The Contractor shall ensure the PM has full authority to act for the Contractor on all matters pertaining to performance of this PWS.

5.6.4.2 The Contractor shall ensure the PM is responsible for ensuring compliance with all local, CCMD, and other requirements that are provided as GFI.

5.6.4.3 Chief Engineer

The Chief Engineer shall lead trade studies, test, evaluation, and integration activities as directed by AGC to maintain the long-term sustainment of AGC's airborne mapping data production. The most promising technologies will be rigorously tested in the laboratory, flight, and operational environments, and, ultimately, may be integrated into the operational systems. The Chief Engineer may lead the integration of existing GFP sensor(s) on new platform(s), integration of advanced sensors into new or existing platforms, and trade studies of the new technologies.

5.6.4.4 Chief Pilot

The Chief Pilot will supervise all aircraft personnel, including pilots, mechanics, technicians, and sensor operators. In coordination with the Program Manager and Site Leads, the Chief Pilot is responsible for defining, communicating and enforcing the policies and procedures related to flight operations as well as the readiness of the personnel while off duty.

5.6.4.5 Chief Mechanic

The Chief Mechanic will supervise and manage all aircraft maintenance, modifications and repair projects on a variety of aircraft in coordination with the Program Manager, Contractor Management and key personnel.

5.6.4.6 Chief Operations Officer

The Chief Operations Officer will supervise all daily operations and will plan and coordinate all ongoing and/or upcoming mobilization and remobilization activities and tasks in

coordination with the Program Manager, Contractor Management and key personnel.

5.6.5 Personnel Management

5.6.5.1 The Contractor shall recruit, vet, prepare, hire, train, replace, and retain personnel to ensure the appropriate background, knowledge, and skills for the performance of this contract are met.

5.6.5.2 The Contractor shall provide for transportation and rotation of personnel.

5.6.5.3 The Contractor shall develop and test procedures for evacuation; notification of next of kin; and continued care for personnel injured, wounded, or killed while performing contract duties and shall apply those procedures as required.

5.6.5.4 The Contractor shall deploy into contingency areas of operation and/or combat zones, when required by the Government, to provide onsite technical expertise, operational support, or training on proper utilization of Government capabilities and services. The Contractor shall comply with all rules and regulations concerning deployment and re-deployment. The Government and Contractor will review the daily threat and intelligence reports for the specific areas of interest we are deployed to as part of the daily operations planning. The Government and Contractor will come to mutual determination whether to conduct operations based on these reports and the associated operations/mission profile.

5.6.5.5 Contractor shall ensure that all deployed personnel are processed through a CRC or similar approved replacement center, as required in accordance with DoD guidance.

5.6.5.6 The Contractor shall ensure employees performing services under this contract are controlled, directed, and supervised at all times by management personnel of the Contractor.

5.6.5.7 The Contractor's management shall ensure that employees properly comply with the performance standards outlined in this PWS and as required by the Contracting Officer or the COR.

5.6.5.8 The Contractor employees shall perform independent of and without the supervision of any Government official.

5.7 Quality

5.7.1 The Contractor shall develop and use a process for testing data quality.

5.7.2 The Contractor shall develop and use data quality and other metrics to assess and demonstrate the effectiveness of the technical approach.

5.7.3 The Contractor shall develop and use a process for collection and analysis of lessons learned.

5.7.4 The Contractor shall use QC and QA procedures to ensure proper oversight for personnel, training, facilities, equipment, supplies, services, and subcontractors.

5.8 Advanced Sensor Analysis, Test and Evaluation

5.8.1 The Contractor shall develop test procedures and success criteria for advanced mapping sensors, as required, or requested from the Government. These procedures shall test for capabilities such as LIDAR gain for high fidelity rooflines and best object resolution; EO resolution in object space; Swath width at 25000 ft AGL; LIDAR auto range gate functionality; Vibration Analysis of LIDAR and EO; Sun angle impact to EO; Ground speed tests; Absolute vertical accuracy; Absolute horizontal accuracy; Non-cardinal direction collection; Race track pattern collection; Noncontiguous/multi-day collection; Determination of minimal acceptable sidelap; Characterize envelope for acceptable AGL; Evaluate use of rapid vs. final PPP for GPS correction information; Perform and evaluate collections in the following region types: urban, desert, heavy foliage, littoral.

5.8.2 The Contractor shall execute the aforementioned tests.

5.8.3 The Contractor shall prepare and deliver a test report. The test report shall contain a remediation plan to correct any critical deficiencies in the sensor, processing or CONOPS that are judged critical to mission success. The Contractor shall deliver an interim test report and a final test report in accordance with the dates specified in the deliverable table of this PWS.

5.8.4 Prior to operational deployment of a new sensor, the Contractor shall collect and process CONUS data at every available opportunity. The contractor shall collect and analyze data to estimate and improve operational readiness, operational availability, Mean Time Between Failure (MTBF) on a configuration item (CI) basis and a LRU basis, Mean Time to Repair (MTTR) on a CI basis and a LRU basis.

5.8.5 The Contractor shall conduct trade studies of new sensors, to include analysis, laboratory tests, flight tests, and data processing tests.

5.9 Sensor Processing Algorithm and Software Development, Enhancement, and Maintenance

5.9.1 The Contractor shall develop new or enhanced sensor processing algorithms and implement them in the processing software.

5.9.2 The Contractor shall enhance and/or maintain the sensor processing software to increase reliability, robustness, performance efficiency, automation levels, product quality, maintainability, or other attributes as identified by AGC.

5.10 Technology Integration

5.10.1 The Contractor shall conduct research of emerging technologies and advancements of LIDAR (topographic and bathymetric), EO, Hyper-spectral and MASINT sensors and integrate these advancements within the 3DH2-M Program.

5.10.2 The Contractor shall provide a detailed cost and implementation plan for emerging sensor technology integration of LIDAR (topographic and bathymetric) and EO on the existing COCO platforms.

5.10.3 The Contractor shall provide a detailed cost and implementation plan for emerging sensor technology integration of Hyper-spectral and MASINT on the existing COCO platforms, and GOCO unmanned platforms.

5.10.4 The Contractor shall fulfill all Army Aviation Engineering Division (AED) Air Worthiness Release (AWR) requirements and receive approved AWRs for all COCO and GOCO aircraft in order to maintain operational capability.

5.11 Value Engineering

In accordance with FAR Part 52.248-1, Value Engineering, it is imperative that the Contractor constantly look for ways to improve services. With technology changing so quickly and providing more and improved features, it is necessary to improve services not only to gain a competitive edge, but in many instances simply to better protect the Warfighter, DoD data and to stay current. "The Contractor is encouraged to develop, prepare and submit value engineering change proposals (VCEPs) voluntarily. The Contractor shall share in any net acquisition savings realized from accepted VECs, in accordance with the incentive sharing rates" of the aforementioned clause.

Part. 6 Attachmentfftechnical Exhibit Listing

Technical Exhibit 1 - Performance Requirements Summary

Technical Exhibit 2 - Deliverables Schedule

Technical Exhibit 3 – Government Property

Technical Exhibit 1 Performance Requirement Summary

The contractor service requirements are summarized into performance objectives that relate directly to mission failure if not accomplished. The performance threshold briefly describes the minimum acceptable levels of service required for each requirement.

PWS Reference	Performance Objective	Standard/Performance Threshold	Method of Surveillance
Quality Control, Quality Assurance PWS 1.5,5.7	PRS#1 Quality Assurance Plan Implements procedures to identify, prevent, and ensure non-recurrence of errors/defects.	Performance consistent with government-approved quality control (QC) program. No more than 10% of errors or defects are attributable to non-compliance with the QC program.	Periodic inspection by the COR.
Manned Fixed Wing Platforms Collection and Processing PWS5.2.2	PRS#2 Data Collection The resolution of the data collected from the manned fixed wing platform is at least one meter for LiDAR and 10 centimeters for EO and an area collection rate of at least 300 square kilometers per flight for the BuckEye I system and 800 square kilometers per flight for the BuckEye II system and 2000 square kilometers per flight for the BuckEye III systems. Collect EO and LiDAR sensor data from xx	For OCONUS collections for the BE I systems and BE II system maintain area collection rate of at least 300 square kilometers per flight, 800 square kilometers per flight for the BuckEye II system and 2000 square kilometers per flight for the BuckEye III systems.	Percentage of data inspected will be determined by the Government. Daily inspection of data collected by OCONUS Government Site Lead/GFR. Weekly inspection by CONUS Government QA Team.
	different areas concurrently. A LiDAR cross line (or cross tie) will be collected during every collection for both CONUS and OCONUS.		COR inspection of processing Logs.
Geospatial Data Processing PWS5.2.2.2	PRS#3 Data Processing - LiDAR Point Clouds LIDAR points (x,y,z,i) in LAS 1.2 format zipped with LASzip. LIDAR points are classified using the LAS specification for ground, low vegetation, medium vegetation, high vegetation, buildings, and water, at a minimum. LIDAR Point Cloud data are tiled as necessary to ensure that the derived LIDAR elevation models are less than 300 megabytes in size. Tiles have 10-20 meters of overlap.	All data media properly marked and controlled. Resolution of the product meets the requirement for the platform. Available no more than 24 hours from the time all the required sensor data are collected. Processing capability located at the flight operations base.	Percentage of data inspected will be determined by the Government. Daily inspection of data processed by OCONUS Government Site Lead to ensure the number of points is commensurate with the target area. Weekly inspection by CONUS Government QA Team.
Geospatial Data Processing PWS5.2.2.2	PRS#4 Data Processing - LiDAR Digital Elevation Model (DEM) Digital representation of the surface topography derived from the LIDAR point cloud files. DEMs are gridded at approximately same resolution as the point cloud data (e.g., point cloud data collected at 1-meter; average point spacing is gridded at 1 meter; point cloud data collected at 50-centimeter average; point spacing gridded at 50 centimeters). The DEM is represented as a raster grid in 32-bit Geospatial Tagged Image File Format (GeoTIFF) gridded at 50 centimeters). The DEM is represented as a raster grid in 32-bit Geospatial Tagged Image File Format (GeoTIFF).	All data media properly marked and controlled. Resolution of the product meets the requirement for the platform. Available no more than 24 hours from the time all the required sensor data are collected. Processing capability located at the flight operations base.	Percentage of data inspected will be determined by the Government. Daily inspection of data processed by OCONUS Government Site Lead to ensure the number of points is commensurate with the target area. Weekly inspection by CONUS Government QA Team.

Geospatial Data Processing PWS5.2.2.2	PRS#5 Data Processing - LiDAR LiDAR Bare Earth DEM Digital representation of bare earth terrain represented as a raster grid in 32-bit GeoTIFF. Bare earth tiles match the DEM and intensity image tiles in extent and resolution.	All data media properly marked and controlled. Resolution of the product meets the requirement for the platform. Available no more than 24 hours from the time all the required sensor data are collected. Processing capability located at the flight operations base.	Percentage of data inspected will be determined by the Government. Daily inspection of data processed by OCONUS Government Site Lead to ensure the number of points is commensurate with the target area. Weekly inspection by CONUS Government QA Team.
Geospatial Data Processing PWS5.2.2.2	PRS#6 Data Processing - LiDAR LiDAR Intensity Image Grayscale LIDAR image depicting reflectivity of the surface topography, derived from the LIDAR Point cloud tiles. Represented as a raster grid in 8-bit GeoTIFF tiles.	All data media properly marked and controlled. Resolution of the product meets the requirement for the platform. Available no more than 24 hours from the time all the required sensor data are collected. Processing capability located at the flight operations base.	Percentage of data inspected will be determined by the Government. Daily inspection of data processed by OCONUS Government Site Lead to ensure the number of points is commensurate with the target area. Weekly inspection by CONUS Government QA Team.
Geospatial Data Processing PWS5.2.2.2	PRS#7 Data Processing - EO Individual Image, Orthophotos, Orthophoto Mosaics EO images run through orthorectification process, resampling the imagery at uniform horizontal ground spacing. Orthophoto mosaics created from individual images. Orthophoto mosaics should be color balanced to reduce frame-to-frame and line-to-line radiometric effects. Individual images in GeoTIFF format. Orthophoto mosaics in GeoTIFF, and Joint Photographic Experts Group (JPEG) 2000--compressed National Imagery Transmission Format (NITF) files.	All data media properly marked and controlled. Resolution of the product meets the requirement for the platform. Available no more than 24 hours from the time all the required sensor data are collected. Processing capability located at the flight operations base.	Percentage of data inspected will be determined by the Government. Daily inspection of data processed by OCONUS Government Site Lead to ensure the number of points is commensurate with the target area. Weekly inspection by CONUS Government QA Team.
Dissemination, Storage, and Maintenance PWS5.2.6	PRS#10 Data Dissemination/Storage	100 % Data Backup, 100% Data Verification Spreadsheets, daily, and weekly reports will reflect the projects completed, storage location, and if a drive is being delivered to AGC. Emails will be sent regarding drive deliveries, drive shipments (will include government), tracking numbers, and drive contents. All data media properly marked and controlled. Data dissemination/delivery time for non-operational customers within 4 weeks . 90% of the time, for operational customers as soon as available 95% of the time.	Percentage of data inspected will be determined by the Government. Daily inspection of data dissemination by Government Site Lead.

Manned Fixed Wing Platforms Collections/Mobilization PWS 5.2.2/5.3	PRS#11 Flight Operations Conduct manned flight operations in 5.2.1 OCONUS locations.	Ensure all flight operations are IAW DCMA 8210 and AR 95-1. MFW flight operations personnel hold current and necessary qualifications. Operational readiness of 90% is maintained.	Periodic inspection by appointed Government Flight Representatives (GFRs) and COR Annual GFR site inspections. All flight and ground operations IAW the Flight and Ground Operating Procedures (FGOPs) and IAW DCMA 8210 and AR 95-1 are subject to inspection.
Manned Fixed Wing Platforms Collections/Mobilization PWS 5.2.2/5.3	PRS#12 Flight Records and Mission Reports Log book record of all flights performed, showing date, take-off time, landing time, duration of flight, time over target (TOT), fuel consumption, names of personnel onboard, and the observations or incidents peculiar to that flight.	All flight records and mission report media properly marked and controlled and delivered monthly in an agreed-on media and format. Area coverage rates meet or exceed minimum	Percentage of data inspected will be determined by the Government Monthly inspection of flight records and mission reports
	Material production and distribution meet the needs of the customer. Delivered on the agreed-on schedule. Report is complete. Achieve 180 km ² /hour for the BE/AC systems, 250 km ² /hour for the BE II system, and 2000 km ² /hour for the BE III systems. Maintain a system operational readiness sufficient for the mission.	required. System operational readiness of at least 90% maintained.	Monthly inspection of flight log by GFR/COR.
Technology Integration PWS5.10	PRS#13 Technology Integration Technology upgrades identified that provide best-in-class capabilities to support objectives	Consistently introduces system upgrades and enhancements within months of public release.	Monthly in-process reviews
Contractor Furnished Items and Responsibilities PWS4.1	PRS#14 Aircraft Maintenance Maintain current airframe and system readiness and logistics issues, upcoming time before overhaul (TBO) replacements, programmed maintenance schedule, and estimated fully mission capable (FMC) dates for inoperable aircraft or systems.	For OCONUS, maintain a maximum of 150 flight hours per month per manned fixed wing platforms. For CONUS collections, maintain a maximum of 25 flight hours per month per manned fixed wing platform. Ensure all flight maintenance IAW DCMA 8210 and AR 95-1, as well as ensure a 90% operational readiness is maintained.	Periodic inspection by the GFR and COR of maintenance logs.
Government Furnished Equipment; Maintain the Airborne LIDAR Systems and Maintain the Airborne EO Systems for the MFW Platforms Collection and Processing; PWS3.3	PRS#15 Sensor Maintenance User availability of equipment meets required operational requirements in PWS.	System operational	System operational

Documentation PWS 1.9	PRS#16 Status reports, Meeting Minutes Contractor shall provide status reports as required by PWS.	Reports are timely, accurate and complete. Passes COR review and delivered on time 90% of the time.	100% inspection by COR
Risk Management PWS5.6.2	PRS #18 Risk Management Risk management program monitors program status to identify and mitigate risks. A risk mitigation plan shall be delivered within three (3) days of identified risks	Changes to risk status identified and reported to COR within one day after	100% inspection by COR

Technical Exhibit 2 - Deliverables Schedule

Number	Deliverable	Reference	Delivery
A001	LIDAR Point Clouds	5.2.2.2 Geospatial Data Processing	As requested by the COR
A002	DEM	5.2.2.2 Geospatial Data Processing	As requested by the COR
A003	Bare Earth DEM	5.2.2.2 Geospatial Data Processing	As requested by the COR
A004	LIDAR Intensity Image	5.2.2.2 Geospatial Data Processing	As requested by the COR
A005	Individual Image Orthophotos	5.2.2.2 Geospatial Data Processing	As requested by the COR
A006	Orthophoto Mosaics	5.2.2.2 Geospatial Data Processing	As requested by the COR
A007	Colorized Point Clouds	5.2.2.2 Geospatial Data Processing	As requested by the COR
A008	Bathymetry products	1.2 Scope	As requested by the COR
A009	Flight Records and Mission Reports	1.15.5 Documentation	As requested by the GFR/COR
A010	Maintenance Report	1.15.5 Documentation	Per Maintenance Event
A011	Monthly Status Reports	1.15.1 Documentation	By 10th day of each calendar month
A012	Weekly Status Reports	1.15.3 Documentation	Weekly
A013	Daily Maintenance Report	1.15.10 Documentation	Daily
A014	Final Report	1.15.6 Documentation	Draft - one (1) month prior to the end of the contract Final - one (1) week prior to the end of the contract
A015	Property Control Plan	1.15.7 Documentation	Within one (1) week after contract award
A016	Transition Plan	1.15.8 Documentation	At proposal submission date and eight (8) months prior to the end of the contract. The Government will give 30 days' notice to when the second transition plan is due.
A017	OPSEC Plan	1.15.9 Documentation	Within ninety (90) calendar days after contract award
A018	Presentation Materials	1.15 Documentation	As requested by the PM/COR
A019	Accidents, Serious Incidents, and Compliance Reports	1.15.5 Documentation	No later than one (1) day after event

A020	Remobilization Plan	5.3.2 Remobilization	Within one (1) week after Government COR or PM notification of remobilization to Contractor. Weekly updates provided during remobilization.
A021	Quality Control Plan	1.5 Quality Control	Within 30 days of contract award
A022	Interim Advanced Sensor Test Report	5.5.4 CONUS Training and Support	Per Testing Event
A023	Final Advanced Sensor Test Report	5.5.4 CONUS Training and Support	Per Testing Event
A024	Sensor Processing Algorithms & Processing Software Enhancements	5.9 Sensor Processing Algorithm and Software Development, Enhancement, and Maintenance	As requested by the PM/COR
A025	Small Business Subcontracting Plan and eSRS Reports	1.15.10 Documentation	Semi-annually
A026	Emerging Sensor Technology Plan	5.10 Technology Integration, 5.11 Value Engineering	As requested by the PM/COR
A027	GFP Master Report and CAP	1.15.7 Documentation	30 Days after contract award. Monthly thereafter

Technical Exhibit 3 – Government Property

The ongoing task orders for this contract include a listing of previously procured Government Property (GP) that the Contractor will have access to and hold stewardship responsibility for. The GFP Attachment (**annotate where it is located**) provides the list of GFP available at the time that the Request for Proposal is issued. Additional GP may be procured under (**annotate task/CLIN**) that will be deemed as CAP but will be managed the same as GP until such time as the Government takes title. The Contractor shall control all GP/CAP in accordance with FAR 52.245-1, 52.245-9, DFARS 252.211-7003, 252.245-7003, 252.242-7005 and the Contractor's Property Management System. Generally, the Contractor shall be responsible for the following, as well as the additional requirements outlined in this section:

- Perform a physical inventory on 10% of all property monthly. Provide the physical inventory in spreadsheet form. These inventories shall be sent to the USACE PM, KO, and Government Property Administrator (GPA).
- Manage the receipt, storage, issue, and transportation of all property to include posting received and issued materials to a web-based system within 24 hours of receiving or issuing, fill, backorder as necessary, close out, and provide status to user-on-user requisitions within 48 hours of receipt of order.
- Update property inventory with items issued. Include date of issue and destination of property/material.
- Manage system access by adding new user accounts, removing accounts no longer required, and assigning appropriate user access levels.
- Maintain system catalog item descriptions, category, images and adding or deleting products as needed and track all orders with expected delivery dates.
- Submit orders through contractor management for review and forward for approval by the Government for items which are not in stock. If substitute items are not acceptable or if requested items are not included in the contract award, the Contractor shall notify contractor management and Government representatives.
- Monitor feedback from users and administrators on website. Report errors or issues to contractor management immediately upon discovery and report all errors to the Government.
- Store property/materials to ensure protection from weather, degradation, and theft
- Track and report usage of materials daily, monthly, and quarterly.
- Maintain (**list how many day stock level**) - the Contractor shall maintain an average monthly usage and then ensure at least a (**list how many day**) supply of any materials is available at all times. Additionally, the material management system shall specify a process of inventorying current stock and turning in excess material to Defense Logistics Agency (DLA).
- Provide a monthly ordering list to the Government to fulfill orders and to maintain the (**list how many days**) stock level.

The Contractor shall provide for the maintenance and repair of all GP/CAP in their possession.

The Contractor shall maintain a purchasing and distribution system at the Administrative Office to provide the GP/CAP to the appropriate personnel and areas.

The Contractor shall establish and implement property management plans, systems, and procedures at the contract, program, site or entity level to enable the outcomes outlined in FAR 52.245-1(f). Failure to maintain an acceptable property management system, as defined in DFARS 252.245-7003, may result in disapproval of the system by the KO and/or withholding of payments.

The Contractor shall process expendable/durable and non-expendable property item requisitions as needed. The Contractor shall review and verify that the requisitions are completed with required information. The

Contractor must have the KO, and the designated representatives sign all property requisitions. The Contractor shall establish a hard copy supporting document file for each document register entry. The Contractor shall process the requisition in accordance with their established Property Management System.

The Contractor shall process receipts of complete and partial shipments and prepare material receiving reports to be provided to the Property Manager and forwarded to the Government Property Administrator and the KO.

The Contractor shall provide property accounting and material management services to include providing accounting for and controlling personal property (Contractor owned property); providing inspection and inventory support; providing special program management as required, and maintenance and repair for GP/CAP.

The Contractor shall maintain up-to-date Government property records for all GP/CAP. All maintenance parts are accountable on the property listing. The Contractor shall maintain a receipt and issue listing, in accordance with FAR Clause 52.245-1. Additionally, any Army issued equipment (if applicable) does not become a part of the Contractor's property accountability report, as they are accountable by the individual that they are issued to. The Contractor, however, shall track these items on a separate listing.

The Contractor shall process lateral transfers and change document lateral transfers as directed and approved by the KO and/or GPA by the use of an SF 30 (Amendment of Solicitation/Modification of Contract).

The Contractor shall process turn-ins of GP/CAP that is excess during the life of this contract through the use of the Government Property disposition tool within the Plant Clearance capability of the Procurement Integrated Enterprise Environment (PIEE) Government Furnished Property (GFP) Module.

The Contractor shall conduct inventories/inspections of GP/CAP as directed by the KO and GPA and IAW FAR Clause 52.245-1.

The Contractor shall perform annual and periodic inventories in accordance with FAR Clause 52.245-1, and the Contractor's Property Management Plan. The Contractor shall post current inventory data to the master property database. Inventory results shall be prepared per FAR Clause 52.245-1. The Contractor shall adjust data elements provided as directed by the KO/GPA.

The Contractor shall report any loss of Government Property through the use of the Government Property Loss tool within Property Loss capability of the Procurement Integrated Enterprise Environment (PIEE) Government Furnished Property (GFP) Module.

The Contractor shall prepare all reports and necessary documents on contract closeout IAW FAR Clause 52.245-1.

End of PWS